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#include <avr/io.h>
#include <util/delay.h>
#include <avr/interrupt.h>
#include <Arduino.h>

// global variables
float distance = 0; //distance of object from ultrasonic sensor
volatile float pulse_duration = 0; //pulse duration measured by timer1
volatile int input_flag = 1;
const float v_sound = 340; // speed of sound wave

// motor functions
// see last lines for details
void drive_forward(void);
void drive_backward(void);
void turn_left(void);
void turn_right(void);
void stop_motors(void);

// QTI sensor functions

// Right QTI (arduino D7)
//return 1 if high (black detected)
int getQTIRight(void) {
    if(PIND & 0b10000000) {
        return 1;
    }
    else{
        return 0;
    }
}

// Left QTI (arduino 6)
// return 1 if high (black detected)
int getQTILeft(void) {
    if(PIND & 0b01000000) {
        return 1;
    }
    else{
        return 0;
    }
}

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    }
}

// US sensor pin change interrupt
// ISR interrupt on D5
ISR(PCINT2_vect){
    if (PIND & 0b00100000){ // if pin 5 is high
        TCNT1 = 0; // timer to 0
    }else{ // pin 5 is low, falling edge of input pulse
        pulse_duration = TCNT1;
        input_flag = 0; //measurement complete
    }
}

// initialize ultrasonic sensor
void initUltrasonic(void){

    PCICR = 0b00000100;
    PCMSK2 = 0b00100000; // enable PCINT23 on pin 5
    TCCR1A = 0b00000000; // normal mode
    TCCR1B = 0b00000010; // timer1 prescaler to 8
}

// get distance from US sensor
float readUltrasonic(void){
    input_flag = 1; //reset flag

    DDRD |= 0b00100000; // D5 output
    PORTD |= 0b00100000; // trigger high

    _delay_us(5); // 5 microsecond pulse

    PORTD &= 0b11011111; // trigger low
    DDRD &= 0b11011111; // D5 input

    TCNT1 = 0;

    // wait for falling edge of pulse to record
    while(input_flag == 1 && TCNT1 < 60000){
    }
}

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    //falling edge not detected --> return large number for no object
    detected
    if(input_flag == 1){
        return 999; // no echo detected
    }

    //return distance from timer count
    return v_sound * pulse_duration * 8 * 62.5e-9 * 100 / 2;
}

// main
int main(void){

    Serial.begin(9600); // initialize serial to debug
    //initColor();
    //getColor();

    DDRB |= 0b00001111; //motor outputs

    // initialize US sensor
    initUltrasonic();

    sei();

    // QTI readings as variables
    volatile int valRight;
    volatile int valLeft;

    //unused --> main script removed before competition
    int mode = 0;
    float obstacleDistance = 15.0;

    //hard code motor functions
    drive_forward();
    _delay_ms(3000); //straight 2 feet

    turn_right();
    _delay_ms(570); // right 90

    drive_forward();

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        _delay_ms(5000);

        turn_right();
        _delay_ms(1150);

        turn_right();
        _delay_ms(570);

        drive_forward();
        _delay_ms(1000);

        stop_motors();

    }

// motor functions
// D8 = forward left motor
// D9 = reverse left motor
// D10 = forward right motor
// D11 = reverse right motor

void drive_forward(void){
    //left and right motor fowards(D8/D10 both high)
    PORTB = (PORTB & 0b11110000) | 0b00000101;
}

void drive_backward(void){
    // left and right motor backwards (D9/D11 both high)
    PORTB = (PORTB & 0b11110000) | 0b00001010;
}

void turn_left(void){
    // left backwards, right forwards (D9 high, D10 high)
    PORTB = (PORTB & 0b11110000) | 0b00000110;
}

void turn_right(void){
    //right backwards, left forwards (D8 and D11 high)
    PORTB = (PORTB & 0b11110000) | 0b00001001;
}

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void stop_motors(void) {  
    //all outputs low --> stop motors  
    PORTB = (PORTB & 0b11110000);  
}
```